Energy-Based Models: Training Without Sampling

CS236: Deep Generative Models — Lecture 12

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Outline

- Recap: EBMs and MCMC sampling
- Today's goal: training without sampling
- Score function and Score Matching
- Noise Contrastive Estimation (NCE)
- NCE for EBMs
- Flow Contrastive Estimation
- Adversarial Training for EBMs
- Comparison and Summary

Recap: EBMs and MCMC

- EBM: $p_\theta(x) = \dfrac{\exp(f_\theta(x))}{Z(\theta)}$
- $Z(\theta)$ intractable $\Rightarrow$ no direct likelihood evaluation
- Ratio $p_\theta(x)/p_\theta(x') = \exp(f_\theta(x) - f_\theta(x'))$ is tractable
- Contrastive Divergence: $\nabla_\theta \log p_\theta \approx \nabla_\theta f_\theta(x_{\text{train}}) - \nabla_\theta f_\theta(x_{\text{sample}})$
- Key bottleneck: need $x_{\text{sample}} \sim p_\theta$ at every training step

MH-MCMC and Langevin MCMC: Limitations

- Metropolis-Hastings: accept/reject based on energy ratio
  - Converges to p_θ in theory via detailed balance
  - In practice: very slow, especially in high dimensions
- Langevin MCMC: $x_{t+1} = x_t + \epsilon \nabla_x f_\theta(x_t) + \sqrt{2\epsilon}\, z_t$
  - Score $\nabla_x \log p_\theta(x) = \nabla_x f_\theta(x)$ requires no $Z(\theta)$
  - Still slow in high dimensions; expensive per training step

Today's Goal: Training Without Sampling

- Contrastive Divergence: need samples from p_θ at each step
- Can we train EBMs without sampling from the model?
- Three approaches:
    - Score Matching: match score functions directly
    - Noise Contrastive Estimation: binary classification vs. noise
    - Adversarial Training: variational upper bound on $\log Z(\theta)$

Score Function

- Score function (Stein score): $s_\theta(x) := \nabla_x \log p_\theta(x)$
- For EBMs: $s_\theta(x) = \nabla_x f_\theta(x) - \underbrace{\nabla_x \log Z(\theta)}_{=0} = \nabla_x f_\theta(x)$
- Score is independent of $Z(\theta)$
- Examples:
  - Gaussian: $s_\theta(x) = -(x-\mu)/\sigma^2$
  - Gamma: $s_\theta(x) = (\alpha - 1)/x - \beta$

Score Matching: Idea

- Fisher divergence between p_{data} and p_θ:
- $D_F(p_{\text{data}}, p_\theta) = \frac{1}{2}\mathbb{E}_{x\sim p_{\text{data}}}[\|\nabla_x \log p_{\text{data}}(x) - s_\theta(x)\|^2]$
- Minimizing D_F = matching score functions
- Both sides evaluated at the same $x \sim p_{\text{data}}$
- Problem: $\nabla_x \log p_{\text{data}}(x)$ is unknown!

Score Matching: Integration by Parts

- Using integration by parts, Fisher divergence simplifies to:
- $D_F = \mathbb{E}_{x \sim p_{\text{data}}} \left[\frac{1}{2} \|\nabla_x \log p_\theta(x)\|^2 + \text{tr}(\nabla_x^2 \log p_\theta(x)) \right] + \text{const}$
- Score matching loss (no p_{data} score needed!):
- $\mathcal{L}_{\text{SM}}(\theta) = \mathbb{E}_{x \sim p_{\text{data}}} \left[\frac{1}{2} \|\nabla_x f_\theta(x)\|^2 + \text{tr}(\nabla_x^2 f_\theta(x)) \right]$
- Caveat: computing $\text{tr}(\nabla_x^2 f_\theta(x))$ (Hessian trace) is expensive

Score Matching: Algorithm and Variants

- Algorithm: no sampling from EBM needed
  - Sample mini-batch $\{x_i\} \sim p_{\text{data}}$
  - Compute loss $\frac{1}{n}\sum_i [\frac{1}{2}\|\nabla_x f_\theta(x_i)\|^2 + \text{tr}(\nabla_x^2 f_\theta(x_i))]$
  - SGD on θ
- Hessian trace bottleneck $\Rightarrow$ efficient variants:
  - Denoising Score Matching (Vincent 2010)
  - Sliced Score Matching (Song et al. 2019)
- These lead naturally to score-based and diffusion models (Lecture 13)

Recap: Two Training Objectives So Far

- KL divergence $\Leftrightarrow$ Maximum Likelihood:
  - Gradient: $\nabla_\theta f_\theta(x_{\text{data}}) - \nabla_\theta f_\theta(x_{\text{sample}})$ (Contrastive Divergence)
  - Requires sampling from p_θ
- Fisher divergence $\Leftrightarrow$ Score Matching:
  - $\mathbb{E}_{x \sim p_{\text{data}}}[\frac{1}{2}\|\nabla_x f_\theta(x)\|^2 + \text{tr}(\nabla_x^2 f_\theta(x))]$
  - No sampling required; but Hessian trace is expensive

Noise Contrastive Estimation (NCE)

- Key idea: distinguish data from a known noise distribution $p_n(x)$
- Train discriminator $D_\theta(x) \in [0, 1]$:
- $\max_\theta \mathbb{E}_{x \sim p_{\text{data}}}[\log D_\theta(x)] + \mathbb{E}_{x \sim p_n}[\log(1 - D_\theta(x))]$
- Optimal discriminator: $D^*(x) = \dfrac{p_{\text{data}}(x)}{p_{\text{data}}(x) + p_n(x)}$
- Parameterize D_θ using model probabilities to recover p_{data}

NCE for EBMs

- Parameterize the discriminator with EBM:
- $D_{\theta,Z}(x) = \dfrac{e^{f_\theta(x)}/Z}{e^{f_\theta(x)}/Z + p_n(x)}$
- Treat Z as an additional trainable scalar (not constrained)
- Optimal solution: $p_{\theta^*,Z^*}(x) = \dfrac{e^{f_{\theta^*}(x)}}{Z^*} = p_{\text{data}}(x)$
- Z^* converges to the true partition function automatically
- NCE objective gives both f_θ *and* an estimate of Z

NCE Training for EBMs

- $\max_{\theta,Z} \mathbb{E}_{x\sim p_{\text{data}}}[\log D_{\theta,Z}(x)] + \mathbb{E}_{x\sim p_n}[\log(1 - D_{\theta,Z}(x))]$
- Equivalently:
- $\max_{\theta,Z} \mathbb{E}_{p_{\text{data}}}[f_\theta(x) - \mathsf{lse}(f_\theta(x), \log Z + \log p_n(x))]$
- $\quad +\mathbb{E}_{p_n}[\log Z + \log p_n(x) - \mathsf{lse}(f_\theta(x), \log Z + \log p_n(x))]$
- Algorithm: sample data + noise, compute NCE loss, SGD on θ and Z
- No sampling from EBM required!

NCE vs. GAN

- Similarities:
  - Both train a binary discriminator with cross-entropy loss
  - Both are likelihood-free
- Differences:
  - GAN: adversarial/minimax training; NCE: no adversarial training
  - NCE: requires $\log p_n(x)$ (tractable noise); GAN: only needs samples from prior
  - NCE: trains an EBM; GAN: trains a deterministic generator

Flow Contrastive Estimation

- NCE requires a good noise distribution $p_n(x)$
  - Noise should be close to p_{data} but tractable
- Flow Contrastive Estimation (Gao et al. 2020):
  - Use a normalizing flow $p_{n,\phi}(x)$ as the noise distribution
  - Train flow to minimize $D_{JS}(p_{\text{data}}, p_{n,\phi})$
  - Joint min-max: $\min_\phi \max_{\theta,Z}$ (flow + EBM together)
- Results: high-quality samples on SVHN, CIFAR-10, CelebA

Adversarial Training for EBMs

- Variational upper bound on $\log Z(\theta)$ using distribution $q_\phi(x)$:
- $\log Z(\theta) = \log \int q_\phi(x) \frac{e^{f_\theta(x)}}{q_\phi(x)} dx \le \mathbb{E}_{x \sim q_\phi}[f_\theta(x)] - H(q_\phi)$
- Substituting into log-likelihood:
- $\mathbb{E}_{p_{\text{data}}}[\log p_\theta(x)] \ge \mathbb{E}_{p_{\text{data}}}[f_\theta(x)] - \mathbb{E}_{q_\phi}[f_\theta(x)] - H(q_\phi)$
- Minimax: $\max_\theta \min_\phi \mathbb{E}_{p_{\text{data}}}[f_\theta(x)] - \mathbb{E}_{q_\phi}[f_\theta(x)] - H(q_\phi)$
- q_ϕ: must support efficient sampling and entropy estimation

Summary

- EBMs are flexible but suffer from intractable $Z(\theta)$
- Three sampling-free training methods:
  - Score Matching: minimize Fisher divergence (match score functions); no $Z(\theta)$ needed; Hessian trace expensive
  - NCE: binary classification vs. noise; learns f_θ and Z simultaneously; needs tractable p_n
  - Adversarial Training: variational lower bound on log-likelihood; minimax with generator q_ϕ
- Next: Score-based and Diffusion Models (Lecture 13)